SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA0605 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptrons, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 **Total Marks:25**

Annexure A

Analytical Problem Solving

Code of Conduct:

I ***THAMIZHARASAN S (192424087)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
THAMIZHARASAN S

Annexure A

Short Answer Test

1, A simple perceptron model is used to classify whether a student passes or fails based on two input features, namely study hours and attendance. The weights associated with these inputs are 0.6 and 0.4 respectively, and the bias value is –0.5. For a given student with study hours equal to 2 and attendance equal to 1, compute the net input to the perceptron and apply the step activation function to determine the final output. Based on the result, conclude whether the student passes or fails. (5 Marks)

Given:

- Input 1 (Study hours), X1=2
- Input 2 (Attendance),X2=1
- Weight 1, w1=0.6
- Weight 2, w2=0.4
- Bias, b=−0.5

Step 1: Calculate the Net Input

The net input to the perceptron is:

$\text{Net Input}=(w1\times x1)+(w2\times x2)+b$

Substitute the given values:

$=(0.6\times 2)+(0.4\times 1)+(-0.5)$

$=1.2+0.4-0.5$

$=1.1$

Net Input = 1.1

Step 2: Apply the Step Activation Function

The step activation function is:

$f(x)=\{1,\text{if } x\geq 0$

$f(x)=\{0,\text{if } x<0$

Step 3: Conclusion

- Perceptron Output: 1
- Interpretation: The student passes.

Final Answer

Net Input = 1.1

- Step Activation Output = 1
- Conclusion: Since the output is 1, the perceptron classifies the student as Pass.

2. In a neural network, a single neuron is trained using the backpropagation learning rule. The initial weight is 0.5 and the learning rate is 0.1. For an input value of 2, the actual output produced by the neuron is 0.6, whereas the target output is 1. Calculate the error in the output and determine the weight update using the delta rule. Hence, compute the new updated weight after one iteration of training. (5 Marks)

Given:

- Initial Weight, $w = 0.5$
- Learning Rate, $\eta = 0.1$
- Input, $x = 2$
- Actual Output, $y = 0.6$
- Target Output, $t = 1$

Step 1: Calculate the Error

The error is:

$\text{Error} = \text{Target Output} - \text{Actual Output}$

$= 1 - 0.6$

$= 0.4$

Error = 0.4

Step 2: Calculate the Weight Update (Delta Rule)

The delta rule is:

$\Delta w = \eta \times \text{Error} \times x$

Substitute the values:

$\Delta w = 0.1 \times 0.4 \times 2$

$= 0.08$

Weight Update (Δw) = 0.08

Step 3: Calculate the New Weight

New Weight=Old Weight+Δw =0.5+0.08

=0.58

New Updated Weight = 0.58

3. A deep learning model is developed for disease prediction and its performance is evaluated using a confusion matrix. The model yields 180 true positives, 150 true negatives, 20 false positives, and 50 false negatives. Using these values, compute the accuracy, precision, sensitivity (recall), and specificity of the model. Based on the calculated results, discuss whether the model is reliable for medical diagnosis, considering the importance of minimizing diagnostic errors. (5 Marks)

Given:

- True Positives (TP) = 180
- True Negatives (TN) = 150
- False Positives (FP) = 20
- False Negatives (FN) = 50

Step 1: Calculate Accuracy

Accuracy=(TP+TN/TP+TN+FP+FN)×100

=(180+150/180+150+20+50)×100

=(330/400)×100

=82.5%

Accuracy = 82.5%

Step 2: Calculate Precision

Precision=(TP/TP+FP)×100

=(180/180+20)×100

=(180/200)×100

$=90\%$

Precision = 90%

Step 3: Calculate Sensitivity (Recall)

Sensitivity$==(TP/TP+FN)\times100$

$=(180/180+50)\times100$

$=(180/230)\times100$

$=78.26\%$

Sensitivity (Recall) = 78.26%

Step 4: Calculate Specificity

Specificity$=(TN/TN+FP)\times100$

$=(150/150+20)\times100$

$=(150/170)\times100$

$=88.24\%$

Specificity = 88.24%

4. During the training of a deep neural network model, the training and validation accuracies are observed over multiple epochs. The training accuracy increases steadily from 60% to 95%, while the validation accuracy initially increases from 58% to 72% but then decreases to 68% in later epochs. Analyse this trend and identify the issue affecting the model after a certain point in training. Explain why this issue occurs in neural networks and suggest two effective techniques to address it. Also, discuss the significance of validation accuracy in evaluating the performance of a learning model. (5 Marks)

Given:

- Training Accuracy: Increased from 60% to 95%
- Validation Accuracy: Increased from 58% to 72%, then decreased to 68%

Analysis

Initially, both the training accuracy and validation accuracy increase, indicating that the neural network is learning meaningful patterns from the training data and improving its prediction performance. The validation accuracy reaches its highest value of 72%, showing that the model performs well on unseen data at this stage.

However, in the later epochs, the training accuracy continues to increase from 60% to 95%, while the validation accuracy decreases from 72% to 68%. This trend indicates that the model has started to overfit the training data. Instead of learning general patterns, the network begins to memorize the training samples, including noise and unnecessary details. As a result, its performance on new or unseen data decreases even though the training accuracy continues to improve.

Techniques to Address Overfitting

1. Early Stopping: Training should be stopped when the validation accuracy reaches its maximum value (72%) before it starts decreasing. This prevents the model from overfitting.
2. Dropout: Randomly deactivating some neurons during training reduces the dependence on specific neurons, improves generalization, and helps the model perform better on unseen data.

Significance of Validation Accuracy

Validation accuracy measures how well the trained model performs on unseen data. It is an important metric for evaluating the model's generalization ability. In this case, although the training accuracy reaches 95%, the validation accuracy decreases to 68%, indicating that the model is no longer improving on new data. Therefore, validation accuracy helps detect overfitting and is used to select the best model.

Conclusion

From the given observations, it is clear that the neural network performs well during the initial training phase, but after the validation accuracy reaches 72%, the model begins to overfit. Therefore, the model should be stopped at the optimal epoch using early stopping, and techniques such as dropout should be applied to improve its performance and reliability on unseen data.

5. A perceptron model is used to classify whether a loan is approved based on two features: income and credit score. The weights are 0.7 and 0.5, and the bias is –0.6. For an applicant with income = 1.5 and credit score = 2, compute the net input and apply the step activation function. Determine whether the loan is approved or rejected. (5 Marks)

Given:

- Income (x1) = 1.5
- Credit Score (x2) = 2
- Weight for Income (w1) = 0.7
- Weight for Credit Score (w2) = 0.5
- Bias (b) = –0.6

Step 1: Calculate the Net Input

The net input to the perceptron is:

$\text{Net Input} = (w1 \times x1) + (w2 \times x2) + b$

Substitute the given values:

Net Input = $(0.7 \times 1.5) + (0.5 \times 2) + (-0.6)$

$= 1.05 + 1.00 - 0.6$

$= 1.45$

Therefore, the Net Input = 1.45

Step 2: Apply the Step Activation Function

The step activation function is:

- If Net Input ≥ 0, Output = 1
- If Net Input < 0, Output = 0

Since the net input is 1.45, which is greater than 0,

Output = 1

Conclusion

The perceptron output is 1, which indicates that the loan is approved. Therefore, based on the given income, credit score, weights, and bias, the applicant qualifies for loan approval.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand